

**National Technical University of Athens**

Department of Electrical and Computer Engineering

# **Modeling Smart Electric Vehicle Charging**

**A Multi-Level Intelligence Approach with  
Renewable Energy Integration**

**Diploma Thesis**

by

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# Περίληψη

Η παρούσα διπλωματική εργασία εξετάζει την έξυπνη φόρτιση ηλεκτρικών οχημάτων σε κτίρια γραφείων με ενσωμάτωση φωτοβολταϊκών, με στόχο να αξιολογήσει πώς διαφορετικά επίπεδα ευφυίας επηρεάζουν το κόστος, τη χρήση της τοπικής ανανεώσιμης ενέργειας και την τήρηση λειτουργικών περιορισμών δικτύου. Η φόρτιση στον χώρο εργασίας αποκτά όλο και μεγαλύτερη σημασία, καθώς η αύξηση των ηλεκτρικών οχημάτων επιβαρύνει τα κτιριακά και δικτυακά συστήματα, ιδιαίτερα όταν η φόρτιση πραγματοποιείται χωρίς κάποιο αυτοματισμό. Τα κτίρια γραφείων αποτελούν κατάλληλο πεδίο εφαρμογής, επειδή συνδυάζουν προβλέψιμα ωράρια εργασίας, μεγάλες περιόδους στάθμευσης και δυνατότητα αξιοποίησης ενέργειας από φωτοβολταϊκά.

Για τον σκοπό αυτό αναπτύχθηκε ένα μοντέλο που περιλαμβάνει το ηλεκτρικό φορτίο του κτιρίου, την παραγωγή από φωτοβολταϊκό σύστημα, αποθήκευση ενέργειας, στόλο ηλεκτρικών οχημάτων και περιορισμούς δικτύου. Η μεθοδολογία βασίζεται σε μελέτη περίπτωσης με ρεαλιστικά δεδομένα κατανάλωσης, ηλιακής ακτινοβολίας και τιμών ηλεκτρικής ενέργειας. Εξετάζονται μια απλή στρατηγική βασισμένη σε κανόνες και μια στρατηγική ενισχυτικής μάθησης με Q-learning, ενώ διερευνάται και η επέκταση σε λειτουργία Vehicle-to-Grid, όπου τα οχήματα μπορούν να επιστρέφουν ενέργεια προς το κτίριο ή το δίκτυο. Η αξιολόγηση πραγματοποιείται με οικονομικά, ενεργειακά, λειτουργικά και υπολογιστικά κριτήρια.

Τα αποτελέσματα δείχνουν ότι η συντονισμένη φόρτιση βελτιώνει τη συνολική λειτουργία του συστήματος σε σχέση με μια απλή ανελαστική στρατηγική. Η προσέγγιση με απλούς κανόνες επιτυγχάνει αξιόπιστη λειτουργία με χαμηλή πολυπλοκότητα και πλήρη συμμόρφωση με τους βασικούς περιορισμούς. Η προσέγγιση με Q-learning προσφέρει επιπλέον βελτίωση στο οικονομικό αποτέλεσμα και καλύτερη διαχείριση της ενέργειας από το δίκτυο, αλλά απαιτεί μεγαλύτερη υπολογιστική προσπάθεια και διαδικασία εκπαίδευσης. Η συμβολή των φωτοβολταϊκών αποδεικνύεται θετική, καθώς μειώνει την εξάρτηση από το δίκτυο και ενισχύει την τοπική αξιοποίηση ανανεώσιμης ενέργειας, αν και το συνολικό όφελος εξαρτάται από το μέγεθος του συστήματος σε σχέση με τη ζήτηση φόρτισης.

Η ανάλυση του Vehicle-to-Grid δείχνει ότι η τεχνική δυνατότητα δεν μεταφράζεται απαραίτητα σε οικονομικό όφελος υπό όλες τις συνθήκες αγοράς. Όταν η διαφορά μεταξύ τιμών αγοράς και πώλησης δεν είναι επαρκής, η αποφόρτιση δεν είναι ελκυστική και τα οφέλη του V2G παραμένουν περιορισμένα. Συνολικά, η εργασία καταλήγει ότι η έξυπνη φόρτιση με ενσωμάτωση φωτοβολταϊκών αποτελεί μια πρακτική και αποτελεσματική λύση για κτίρια γραφείων, ενώ οι

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πιο προηγμένες στρατηγικές ελέγχου και το V2G αποκτούν αξία κυρίως όταν υποστηρίζονται από κατάλληλες τιμολογιακές και λειτουργικές συνθήκες.

**Λέξεις κλειδιά:** Ηλεκτρικά οχήματα, έξυπνη φόρτιση, ανανεώσιμες πηγές ενέργειας, βελτιστοποίηση, Vehicle-to-Grid, κτίρια γραφείων

# Abstract

This thesis examines the smart charging of electric vehicles (EVs) in office buildings with integrated photovoltaics (PV), aiming to evaluate how different levels of intelligence affect costs, the utilization of local renewable energy, and adherence to grid operational constraints. Workplace charging is becoming increasingly important, as the growth of EVs places a burden on building and grid systems, particularly when charging occurs without automation. Office buildings serve as an ideal field of application because they combine predictable working hours, long parking durations, and the potential to utilize energy from PV systems.

To this end, a model was developed that includes the building’s electrical load, PV system generation, energy storage, an EV fleet, and grid constraints. The methodology is based on a case study using realistic consumption, solar radiation, and electricity price data. A simple rule-based strategy and a reinforcement learning strategy using Q-learning are examined, while the expansion into Vehicle-to-Grid (V2G) operation—where vehicles can return energy to the building or the grid—is also explored. The evaluation is conducted based on economic, energetic, operational, and computational criteria.

The results indicate that coordinated charging improves overall system performance compared to a simple inelastic strategy. The rule-based approach achieves reliable operation with low complexity and full compliance with basic constraints. The Q-learning approach offers further improvement in economic outcomes and better management of grid energy but requires greater computational effort and a training process. The contribution of photovoltaics proves positive, as it reduces grid dependence and enhances local renewable energy utilization, although the overall benefit depends on the system’s size relative to charging demand.

The analysis of Vehicle-to-Grid shows that technical capability does not necessarily translate into economic benefit under all market conditions. When the price spread between buying and selling is insufficient, discharging is unattractive, and V2G benefits remain limited. Overall, the thesis concludes that smart charging with PV integration is a practical and effective solution for office buildings, while more advanced control strategies and V2G gain value primarily when supported by appropriate pricing and operational conditions.

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**Keywords:** Electric vehicles, smart charging, renewable energy, optimization, Vehicle-to-Grid, office buildings

# Acknowledgments

[Write your acknowledgments here]

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# Chapter 1

## Introduction

### 1.1 Background and Motivation

The global transition toward sustainable transportation has accelerated the adoption of electric vehicles (EVs). In 2023, global electric car sales neared 14 million and reached about 18% of all cars sold, showing how quickly EVs are moving into the mass market [iea\\_evo\\_2024](#). As EV penetration grows, workplace charging becomes more important because it makes electric driving more convenient and affordable for employees and supports broader adoption [doe\\_afdc\\_workplace](#). At the same time, uncontrolled charging can stress the grid during peak hours, increase electricity costs, and leave on-site renewable energy underused. These problems are most visible in office buildings, where many vehicles connect at similar times and the electrical capacity is limited.

Smart charging systems address these issues by coordinating EV charging with building energy use, renewable generation, and time-varying electricity prices. Office buildings are a natural setting because parking times are long, usage patterns are regular, and rooftop solar is common. Solar PV is also increasingly cost-competitive, making local generation a practical part of workplace charging strategies [irena\\_rpgc\\_2023](#).

### 1.2 Smart Charging Levels

This thesis investigates multiple levels of charging intelligence. Each level adds coordination or control to the previous one, so we can compare benefits against added complexity. In this structure, the simple rule-based approach is the zero level.

#### 1.2.1 Level 0: Simple Rule-Based Charging

Level 0 uses basic rules to guide charging. The rules are simple, such as using solar surplus first, honoring grid capacity limits, and allocating power in a predictable or-

der. This approach is easy to run and yields reasonable performance without heavy computation.

### **1.2.2 Level 1: Intelligent Optimization**

Level 1 uses optimization to compute better schedules. In this thesis, the main example is reinforcement learning, where a Q-learning agent improves charging decisions through trial and error. This level offers better performance but requires training and more complex implementation.

### **1.2.3 Level 2: Vehicle-to-Grid (V2G)**

Level 3 adds bidirectional energy flow so EVs can discharge back to the building or grid during high-price periods. This enables potential revenue and grid services, but it also adds infrastructure cost and constraints to protect battery health and mobility needs.

## **1.3 Role of Renewable Energy**

Solar photovoltaic systems on office buildings can reduce charging costs and grid dependence. Solar production often peaks around midday, which aligns with typical workplace parking, so EVs can absorb on-site generation when it is available. This reduces grid purchases and helps lower peak demand, while also cutting emissions. Solar PV has become one of the lowest-cost new power options, which strengthens the case for integrating it with charging [\*\*irena\\_rpgc\\_2023\*\*](#). The main challenge is to coordinate charging with variable solar output while still meeting mobility needs and grid limits.

## **1.4 Benefits of Smart Charging in Office Buildings**

Office buildings are a strong fit for smart charging because arrival and departure times are predictable, parking durations are long enough to shift charging, and solar production typically overlaps with working hours. A single building operator can manage both the facility and the chargers, which simplifies control and makes cost savings more direct.

## 1.5 Literature Review

### 1.5.1 Smart EV Charging

Research on smart EV charging shows that coordinated scheduling can reduce costs and avoid grid stress compared with uncontrolled charging. Early studies such as Richardson et al. **richardson2012electric** and Sun et al. **sun2015optimal** formalized cost-minimizing schedules under grid constraints. Later work also explored decentralized and valley-filling strategies to manage peak demand while meeting user needs **zhang2018smart**.

### 1.5.2 Optimization Algorithms

Optimization for charging ranges from classical methods to learning-based control. Reinforcement learning is attractive because it can handle uncertainty and learn policies directly from interaction. Sutton and Barto provide the foundational framework **sutton2018reinforcement**, while applications to energy management and EV charging highlight its practical use **vandael2013reinforcement**. Deep RL further expands the approach by handling larger state spaces **mnih2015human**.

### 1.5.3 Vehicle-to-Grid Technology

V2G technology enables bidirectional energy flow so EVs can provide grid services or earn revenue during high-price periods. Kempton and Tomić established the basic revenue and capacity framework **kempton2005vehicle**, and later studies examined control and aggregation strategies for services such as frequency regulation **han2010development**. Reviews of V2G and related vehicle-to-x concepts highlight both the opportunity and the remaining technical and market barriers **liu2019opportunities**.

### 1.5.4 Renewable Integration

Renewable integration studies show that combining solar PV with smart charging can improve self-consumption and reduce grid draw. An NREL analysis of commercial EV charging highlights that co-deploying PV and storage with charging infrastructure can lower utility costs when charging is coordinated with on-site generation **nrel\_ev\_pv\_2019**.

## 1.6 Research Objectives

The objectives are to develop models for intelligence levels 0–2, implement and compare a simple rule-based strategy and a Q-learning approach, quantify costs, renewable

utilization, and constraint satisfaction, evaluate the added value of V2G, and provide practical guidance on which intelligence level offers the best cost–benefit balance for office buildings.

### **1.7 Thesis Structure**

The remainder of this thesis is organized as follows. Chapter 2 presents the models and algorithms for each intelligence level. Chapter 3 describes the office building case study, data sources, and simulation setup. Chapter 4 reports the results and comparative analysis. Chapter 5 summarizes the findings and outlines recommendations and future work.

# Chapter 2

## Methodology

This chapter describes the model and algorithm for the different parts of the System that is responsible for the EV charging. It includes components, optimizations and implementation for the different optimization levels.

### 2.1 System Components

#### 2.1.1 Office Building Model

The office building is modeled by its energy consumption and its renewable energy generation.

##### Energy Consumption

The building's hourly energy consumption profile  $C_b(t)$  is derived from the ELMAS dataset (European Load Model for Aggregated Sectors), specifically the Office cluster (#5) from `Time_series_18_clusters.csv`.

Let  $L_t$  be the hourly energy load of the Office cluster base series ( $T = 8,736$  hours for year 2018) with yearly energy:

$$E_{cluster} = \sum_{t=1}^T L_t \quad (2.1)$$

To represent a single office building, the per-customer annual energy is calculated for each class  $c$  belonging to the Office cluster column:

$$e_c = \frac{E_c}{N_c} \quad (2.2)$$

where  $E_c$  is the annual energy and  $N_c$  the number of customers in class  $c$ . The required annual energy is set to the 75th percentile of the customer distribution:

$$E_{target} = Q_{0.75}(\{e_c\}) = 58,831 \text{ kWh} \quad (2.3)$$

The energy profile is then scaled to match this target:

$$L_t^{scaled} = L_t \cdot \frac{E_{target}}{E_{cluster}} \quad (2.4)$$

To introduce noise to the hourly energy profile, two multiplicative noise factors are applied—a daily factor  $\alpha_d$  (constant within each day) and an hourly factor  $\beta_t$ :

$$\alpha_d \sim \mathcal{N}(1, \sigma_d^2), \quad \sigma_d = 0.08, \quad \alpha_d \in [0.7, 1.3] \quad (2.5)$$

$$\beta_t \sim \mathcal{N}(1, \sigma_h^2), \quad \sigma_h = 0.03, \quad \beta_t \in [0.9, 1.1] \quad (2.6)$$

The synthetic series before re-normalization:

$$L_t^{syn} = L_t^{scaled} \cdot \alpha_d(t) \cdot \beta_t \quad (2.7)$$

Finally, the series is re-scaled to exactly preserve the target annual energy:

$$C_b(t) = L_t^{syn} \cdot \frac{E_{target}}{\sum_{t=1}^T L_t^{syn}} \quad (2.8)$$

The single-office profile has a mean load of 6.73 kW and a peak load of 11.92 kW.

### Solar Photovoltaic System

The solar PV system creates electricity based on irradiance following the equation:

$$P_{solar}(t) = A_{panel} \cdot \eta_{panel} \cdot I(t) \quad (2.9)$$

where:

- $A_{panel}$  = solar panel area ( $\text{m}^2$ )
- $\eta_{panel}$  = panel efficiency
- $I(t)$  = solar irradiance at hour  $t$  ( $\text{W}/\text{m}^2$ )

### Building Battery Storage

The building has a battery with capacity  $C$  and SOC  $\text{SoC}_b(t)$ .

The max discharge power at time  $t$  is bounded by both the efficiency and the available energy above the minimum SoC:

$$P_{dis,t} = \min(P_{max}, (\text{SoC}_{t-1} - \text{SoC}_{min}) \cdot \eta) \quad (2.10)$$

Similarly, the max charge is bounded by the efficiency and the remaining capacity:

$$P_{ch,t} = \min(P_{max}, C - \text{SoC}_{t-1}) \quad (2.11)$$

The state of charge depends on whether the building has extra or missing energy. When extra solar energy is available, the battery charges:

$$\text{SoC}_t = \text{SoC}_{t-1} + \min(P_{ch,t}, E_{excess}(t)) \quad (2.12)$$

When the building demand overcomes the supply, the battery discharges:

$$\text{SoC}_t = \text{SoC}_{t-1} - \min\left(\frac{P_{dis,t}}{\eta}, \frac{E_{missing}(t)}{\eta}\right) \quad (2.13)$$

where:

- $P_{max}$  = maximum charge/discharge power [kW]
- $C$  = battery capacity [kWh]
- $\text{SoC}_{min}$  = minimum allowable state of charge
- $\eta$  = round-trip efficiency
- $E_{excess}(t) = \max(0, P_{solar}(t) - C_b(t))$  = surplus solar energy
- $E_{missing}(t) = \max(0, C_b(t) - P_{solar}(t))$  = energy deficit

Constraints:

$$\text{SoC}_{min} \leq \text{SoC}_t \leq C \quad (2.14)$$

$$0 \leq P_{ch,t} \leq P_{max} \quad (2.15)$$

$$0 \leq P_{dis,t} \leq P_{max} \quad (2.16)$$

## Net Energy Demand

The building's net energy demand (positive = needs grid, negative = surplus):

$$E_{net}^{building}(t) = C_b(t) - P_{solar}(t) - E_{discharge}^b(t) + E_{charge}^b(t) \quad (2.17)$$

### 2.1.2 Electric Vehicle Fleet Model

#### Individual EV Characteristics

Each EV  $i$  in the fleet has the following info:

- Battery capacity:  $B_{cap}^i$  [kWh]

- State of charge:  $\text{SoC}^i(t)$  [dimensionless, 0-1]
- Maximum charge rate:  $P_{max,charge}^i$  [kW]
- Maximum discharge rate:  $P_{max,discharge}^i$  [kW] (for V2G)
- Arrival time:  $t_{arrival}^i$  [hour]
- Departure time:  $t_{target}^i$  [hour]
- Desired SoC:  $\text{SoC}_{desired}^i$  [dimensionless]

### EV Battery Dynamics

The state of charge evolves for EV  $i$  following the equation:

$$\text{SoC}^i(t+1) = \text{SoC}^i(t) + \frac{\eta_{charge} \cdot E_{charge}^i(t) - E_{discharge}^i(t)/\eta_{discharge}}{B_{cap}^i} \quad (2.18)$$

where  $\eta_{charge}$  and  $\eta_{discharge}$  are charging and discharging efficiencies (typically 0.95).

### Constraints

Each EV must satisfy:

$$\text{SoC}_{min} \leq \text{SoC}^i(t) \leq 1.0 \quad \forall t \quad (2.19)$$

$$\text{SoC}^i(t_{target}^i) \geq \text{SoC}_{desired}^i \quad (2.20)$$

$$E_{charge}^i(t) = 0 \quad \text{if } t < t_{arrival}^i \text{ or } t \geq t_{target}^i \quad (2.21)$$

Constraint (??) makes sure that each EV reaches its desired charge level before departure.

### 2.1.3 Grid Model

#### Dynamic Pricing

The grid uses daily pricing following the equation:

$$\pi_{buy}(t) = \text{price\_profile}[t \bmod 24] \quad [\text{€/kWh}] \quad (2.22)$$

For V2G scenarios, the sell price:

$$\pi_{sell}(t) = \text{sell\_price\_profile}[t \bmod 24] \quad [\text{€/kWh}] \quad (2.23)$$

Typically,  $\pi_{sell}(t) \leq \pi_{buy}(t)$  (80-90% of purchase price).

### Grid Capacity Constraint

The total grid power consumption has a constraint:

$$E_{grid}^{building}(t) + \sum_{i=1}^{N_{EV}} E_{grid}^i(t) \leq P_{grid,max}(t) \quad (2.24)$$

where  $P_{grid,max}(t)$  represents the hourly grid capacity limit.

## 2.2 Optimization Problem Formulation

### 2.2.1 Objective Function

The goal is to minimize the total daily electricity cost:

$$\min \sum_{t=0}^{T-1} [\pi_{buy}(t) \cdot E_{grid}^{total}(t) - \pi_{sell}(t) \cdot E_{V2G}^{total}(t)] \quad (2.25)$$

where:

- $E_{grid}^{total}(t)$  = total energy purchased from grid at hour  $t$
- $E_{V2G}^{total}(t)$  = total energy sold back to grid (V2G only)
- $T$  = simulation duration (24 hours)

### 2.2.2 Energy Balance

For each hour  $t$ , the energy equation must be satisfied:

$$C_b(t) + \sum_{i=1}^{N_{EV}} E_{charge}^i(t) = P_{solar}(t) + E_{grid}^{total}(t) + \sum_{i=1}^{N_{EV}} E_{discharge}^i(t) \quad (2.26)$$

## 2.3 Level 0: Baseline (Uncontrolled Charging)

### 2.3.1 Algorithm Description

In the baseline scenario, each EV charges at max rate upon arrival:

$$E_{charge}^i(t) = \begin{cases} P_{max,charge}^i & \text{if } t_{arrival}^i \leq t < t_{target}^i \text{ and } SoC^i(t) < 1.0 \\ 0 & \text{otherwise} \end{cases} \quad (2.27)$$

### 2.3.2 Mathematical Formulation

The total cost for charging:

$$Cost_{Level0} = \sum_{t=0}^{T-1} \pi_{buy}(t) \cdot \left[ \max(0, E_{net}^{building}(t)) + \sum_{i=1}^{N_{EV}} E_{charge}^i(t) \right] \quad (2.28)$$

#### Key Assumptions:

- No coordination between EVs
- Solar energy used for building first, remainder for EVs if available

## 2.4 Level 1: Simple Rule-Based Charging

### 2.4.1 Algorithm Description

The simple algorithm follows a priority-based heuristic:

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#### Algorithm 1 Simple Rule-Based Charging

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```

1: for each hour  $t$  do
2:   Calculate building surplus:  $E_{surplus}(t) = \max(0, P_{solar}(t) - C_b(t))$ 
3:   Calculate available grid capacity:  $P_{available}(t) = P_{grid,max}(t) - \max(0, C_b(t) - P_{solar}(t))$ 
4:    $E_{remaining} \leftarrow E_{surplus}(t)$ 
5:   for each EV  $i$  (in order) do
6:     if  $t_{arrival}^i \leq t < t_{target}^i$  and  $SoC^i(t) < SoC_{desired}^i$  then
7:        $E_{needed}^i = \min(P_{max,charge}^i, (SoC_{desired}^i - SoC^i(t)) \cdot B_{cap}^i)$ 
8:        $E_{from\_surplus} = \min(E_{needed}^i, E_{remaining})$ 
9:        $E_{from\_grid} = \min(E_{needed}^i - E_{from\_surplus}, P_{available}(t))$ 
10:       $E_{total} = E_{from\_surplus} + E_{from\_grid}$ 
11:      Charge EV  $i$  with  $E_{total}$ 
12:      Update  $E_{remaining}$  and  $P_{available}(t)$ 
13:    end if
14:  end for
15: end for

```

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### 2.4.2 Mathematical Formulation

Energy for EV  $i$  at hour  $t$ :

$$E_{charge}^i(t) = \min(P_{max,charge}^i, (SoC_{desired}^i - SoC^i(t)) \cdot B_{cap}^i, E_{available}(t)) \quad (2.29)$$

where  $E_{available}(t)$  contains both building extra energy and remaining grid capacity.

## 2.5 Level 2: Intelligent Optimization Algorithms

### 2.5.1 Reinforcement Learning (Q-Learning)

#### State Space

The state representation for EV  $i$  at hour  $t$ :

$$s^i(t) = (\text{EV\_id}, \text{SoC}_{bin}^i(t), t, \text{grid\_availability}) \quad (2.30)$$

where  $\text{SoC}_{bin}^i(t)$  is the discretized SoC (bins of 0.1).

#### Action Space

For charge-only systems:

$$\mathcal{A} = \{\text{charge}, \text{standby}\} \quad (2.31)$$

For V2G-enabled systems:

$$\mathcal{A}_{V2G} = \{\text{charge}, \text{discharge}, \text{standby}\} \quad (2.32)$$

#### Reward Function

The reward function is the following:

$$\begin{aligned} R(s, a, s') = & -\alpha_1 \cdot E_{grid}^i(t) \cdot \pi_{buy}(t) \quad (\text{minimize grid cost}) \\ & + \alpha_2 \cdot E_{solar}^i(t) \quad (\text{maximize solar usage}) \\ & + \alpha_3 \cdot \mathbb{K}[\text{SoC}^i(t_{target}^i) \geq \text{SoC}_{desired}^i] \quad (\text{target achievement}) \\ & - \alpha_4 \cdot \max(0, \text{SoC}_{desired}^i - \text{SoC}^i(t_{target}^i)) \quad (\text{penalty for shortfall}) \\ & - \alpha_5 \cdot \mathbb{K}[\text{SoC}^i(t) < \text{SoC}_{min}] \quad (\text{constraint violation}) \end{aligned} \quad (2.33)$$

where  $\alpha_1, \dots, \alpha_5$  are weights.

#### Q-Learning Update Rule

The Q-value is updated using the temporal difference method:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[ R(s, a, s') + \gamma \max_{a'} Q(s', a') - Q(s, a) \right] \quad (2.34)$$

where:

- $\alpha$  = learning rate (0.1-0.15)
- $\gamma$  = discount factor (0.95-1.0)

- $s'$  = next state after taking action  $a$  in state  $s$

### Policy Execution

After training, the policy selects actions greedily:

$$a^*(s) = \arg \max_{a \in \mathcal{A}} Q(s, a) \quad (2.35)$$

## 2.6 Level 3: Vehicle-to-Grid (V2G)

### 2.6.1 Bidirectional Energy Flow

V2G expands the optimization to include discharge decisions:

$$E_{net}^i(t) = E_{charge}^i(t) - E_{discharge}^i(t) \quad (2.36)$$

### 2.6.2 V2G Constraints

Additional constraints for V2G are the following:

$$E_{discharge}^i(t) \leq (\text{SoC}^i(t) - \text{SoC}_{desired}^i) \cdot B_{cap}^i \quad (\text{preserve target SoC}) \quad (2.37)$$

$$E_{charge}^i(t) \cdot E_{discharge}^i(t) = 0 \quad (\text{no simultaneous charge/discharge}) \quad (2.38)$$

### 2.6.3 Revenue Calculation

Total benefit with V2G:

$$\text{Benefit}_{V2G} = - \sum_{t=0}^{T-1} \left[ \pi_{buy}(t) \cdot E_{grid}^{total}(t) - \pi_{sell}(t) \cdot \sum_{i=1}^{N_{EV}} E_{discharge}^i(t) \right] \quad (2.39)$$

Positive benefit indicates revenue or cost saving.

### 2.6.4 V2G Integration with Algorithms

The Q-Learning algorithm is extended to handle V2G by:

1. Expanding action space to include discharge
2. Modifying reward/cost functions to include sell reward
3. Adding discharge constraints to preserve battery state

## 2.7 Key Assumptions and Limitations

### 2.7.1 Assumptions

1. **Perfect Forecasting:** Solar production and building consumption are known in advance
2. **Deterministic Arrivals:** EV arrival/departure times are fixed (within random windows)
3. **No Battery Degradation:** Charging/discharging cycles do not degrade battery capacity
4. **Fixed Efficiency:** Battery charging efficiency is constant at 95%
5. **Ideal Grid Connection:** No transmission losses or power quality issues
6. **Price-Taker:** EV charging does not affect electricity prices

### 2.7.2 Limitations

1. Simplified battery model (linear SoC dynamics)
2. No modeling of battery thermal effects
3. Single-day simulation
4. No consideration of user preferences or comfort
5. Computational complexity and time not analyzed in detail

# Chapter 3

## Case Study: Office Building Scenario

This chapter describes the office building scenario, that is used to evaluate the smart charging algorithms. Data sources are provided along with parameter selection and justification for the chosen values.

The case study simulates a typical office building with a rooftop solar PV installation, EV charging infrastructure for 15 vehicles, a grid connection with capacity constraints, and a 24-hour simulation period.

### 3.1 Building Energy Profile

#### 3.1.1 Energy Consumption Data

##### Data Source

The building energy consumption profile is derived from the ELMAS (European Load Model for Aggregated Sectors) dataset, specifically the Office cluster (#5). The dataset file is `office_single_building_timeseries.csv` and the source is the ELMAS file `Time_series_18_clusters.csv` for Office cluster #5. The building type is office/commercial, the target annual energy is 58,831 kWh (the 75th percentile of the per-customer distribution), and the temporal resolution is hourly with 8,736 data points for year 2018.

The profile was created by scaling the cluster load shape to a single-building target energy and adding stochastic variability (daily  $\sigma_d = 0.08$ , hourly  $\sigma_h = 0.03$ ) to introduce realistic fluctuations, followed by re-normalization to preserve the annual energy total (see Section ?? for full methodology).

## Profile Characteristics

Figure ?? shows the hourly consumption pattern:

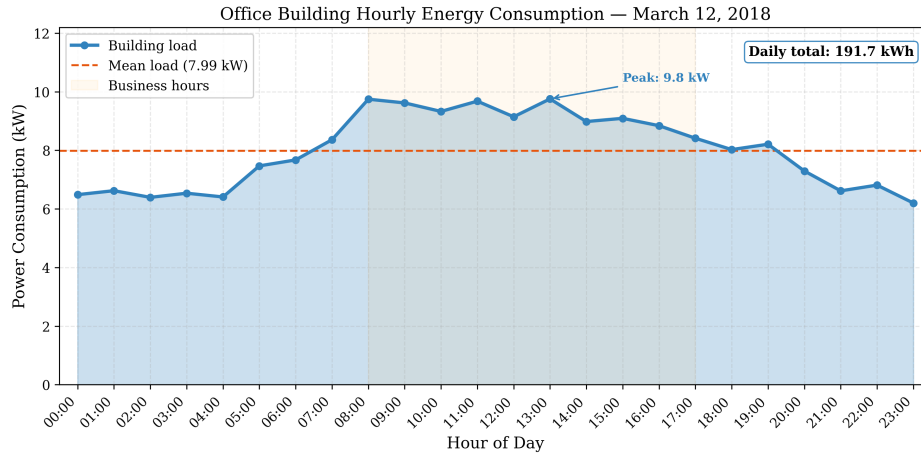


Figure 3.1: Office building hourly energy consumption profile

Annual statistics are a mean load of 6.73 kW, a peak load of 11.92 kW during business hours, an annual energy of 58,831 kWh, and a daily consumption of about 161 kWh (annual divided by 365).

## Validation

The consumption profile comes from the ELMAS source data. The per-customer annual energy distribution, with mean 10,429 kWh and  $P75 = 58,831$  kWh, reflects small to medium-sized office buildings. The 75th percentile is used to represent a medium-to-large office. The cluster load provides realistic hourly patterns with a working-hours peak and an overnight baseline, and stochastic noise is added so the profile is not artificially smooth while preserving the correct total energy.

### 3.1.2 Solar PV System

#### System Specifications

The rooftop solar PV system parameters:

Table 3.1: Solar PV system specifications

| Parameter         | Value                                            |
|-------------------|--------------------------------------------------|
| Panel area        | 150 m <sup>2</sup>                               |
| Panel efficiency  | 20%                                              |
| Peak power        | 30 kW                                            |
| Technology        | Monocrystalline PERC (JA Solar DeepBlue 3.0 Pro) |
| Reference module  | JAM54S31-405/MR (405 Wp, 20.7% efficiency)       |
| Number of modules | 74                                               |

The reference module is the JA Solar DeepBlue 3.0 Pro (JAM54S31-405/MR) `ja_solar_deepblue` a widely deployed monocrystalline PERC half-cut cell panel rated at 405 Wp with a module efficiency of 20.7% (STC). The installation comprises 74 units, yielding a combined unit area of approximately 144.5 m<sup>2</sup> ( $\approx 150$  m<sup>2</sup>) and a peak system power of 29.97 kW ( $\approx 30$  kW).

### Irradiance Data

Solar irradiance data is obtained from PVGIS (Photovoltaic Geographical Information System) `pvgis`. The source is the European Commission Joint Research Centre PVGIS database, using the Paris, France location (48.857°N, 2.352°E, elevation 32 m) and the PVGIS-SARAH3 radiation database. The panel orientation is south-facing (azimuth 0°) with a tilt of 37° (site-optimum). The date is March 12, 2018 as a representative spring day. The temporal resolution is hourly and the data file is `irradiance_pvgis.csv`.

### Solar Production Profile

The hourly solar power production:

$$P_{solar}(t) = A_{panel} \cdot \eta_{panel} \cdot I(t) = 150 \text{ m}^2 \cdot 0.20 \cdot I(t) \quad (3.1)$$

Figure ?? illustrates the production profile:

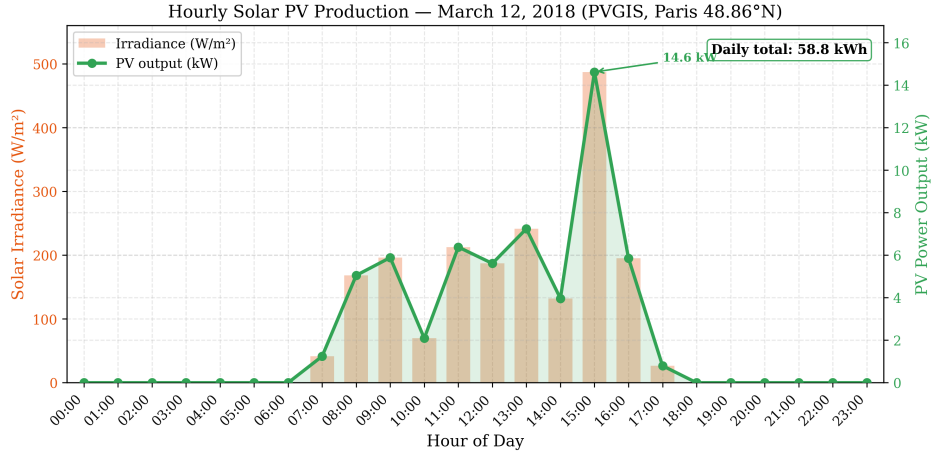


Figure 3.2: Hourly solar PV production profile (March 12, 2018)

Peak production is about 14.6 kW at 15:00, with a peak irradiance of 487 W/m<sup>2</sup>. Total daily production is 58.8 kWh. Solar production partially offsets building consumption during business hours (7:00–17:00), with a brief surplus at 15:00. The profile aligns with EV parking hours (6 AM–6 PM) and shows variable irradiance due to sporadic cloud cover typical of Paris in spring.

### Selection Justification

March 12, 2018 was selected because it is a spring day with moderate solar production of about 59 kWh. It has realistic variable-cloud conditions typical of the Paris region, which provides a test of solar integration without unrealistically clear-sky irradiance. It also represents a typical seasonal pattern rather than extreme summer or winter conditions, and validated satellite-derived irradiance data are available from PVGIS (SARAH3).

### 3.1.3 Building Battery Storage

The building includes a stationary battery storage system:

Table 3.2: Building battery specifications

| Parameter                | Value                        |
|--------------------------|------------------------------|
| Capacity                 | 80 kWh                       |
| Efficiency               | 95%                          |
| Initial SoC              | 50%                          |
| Depth of discharge limit | 80%                          |
| Max charge rate          | 50 kW                        |
| Max discharge rate       | 50 kW                        |
| Chemistry                | Lithium Iron Phosphate (LFP) |
| Reference product        | BYD Battery-Box Premium HVM  |

The battery parameters are representative of a modular Lithium Iron Phosphate (LFP) system based on the BYD Battery-Box Premium HVM series `byd_batterybox`. The HVM system uses cobalt-free LFP cells with a certified round-trip efficiency of  $\geq 96\%$  (HTW Berlin) and offers modular scalability in 2.76 kWh increments. In the simulated configuration, the equivalent of four HVM 19.3 towers ( $4 \times 19.32 = 77.3$  kWh  $\approx 80$  kWh) provides a combined maximum charge/discharge power of approximately 50 kW.

## 3.2 Electric Vehicle Fleet

### 3.2.1 Fleet Composition

The simulation supports a fleet of 15 electric vehicles, representing a typical office building with 50-100 employees (assuming 15-30% EV adoption).

#### Vehicle Specifications

Each EV is modeled with identical technical characteristics:

Table 3.3: Electric vehicle specifications

| Parameter                | Value                      |
|--------------------------|----------------------------|
| Battery capacity         | 100 kWh                    |
| Reference vehicle        | Tesla Model 3 Long Range   |
| Max charge rate          | 22 kW (Level 2 AC charger) |
| Max discharge rate       | 22 kW (V2G capable)        |
| Energy consumption       | 0.18 kWh/km                |
| Depth of discharge limit | 90%                        |
| Minimum SoC              | 20%                        |

#### Justification

The 100 kWh capacity is representative of modern long-range EVs such as the Tesla Model 3 and Hyundai Ioniq 5. A 22 kW charge rate reflects standard Level 2 AC charging that is common in workplace installations. An energy use of 0.18 kWh/km is typical for mid-size EVs, and a 20% minimum SoC prevents deep discharge and limits battery degradation.

### 3.2.2 Mobility Patterns

#### Arrival and Departure Times

The EV fleet follows typical office worker patterns:

Table 3.4: EV mobility parameters

| Parameter         | Value          |
|-------------------|----------------|
| Arrival window    | 6:00 - 9:00 AM |
| Departure window  | 4:00 - 8:00 PM |
| Parking duration  | 7-14 hours     |
| Initial SoC range | 20-30%         |
| Desired SoC range | 70-80%         |

Each EV's specific arrival and departure times are randomly sampled from the specified windows to simulate realistic variability.

### State of Charge Requirements

The initial SoC of 20–30% represents battery level after a morning commute of about 30–50 km. The desired SoC of 70–80% ensures enough charge for the evening commute plus a reserve of roughly 100–150 km.

### Justification

The mobility pattern reflects typical office workers. The morning arrival spread from 6–9 AM accounts for flexible work hours, while the evening departure spread from 4–8 PM reflects varied schedules. Long parking durations provide flexibility for smart charging, and the SoC requirements ensure mobility needs are met.

## 3.3 Grid Parameters

### 3.3.1 Electricity Pricing

#### Day-Ahead Market Prices

The simulation uses real wholesale day-ahead electricity prices from the EPEX SPOT exchange for the French bidding zone on March 12, 2018. The data was gathered from the ENTSO-E Transparency Platform `entsoe_transparency` and the Fraunhofer ISE Energy-Charts API `energy_charts`.

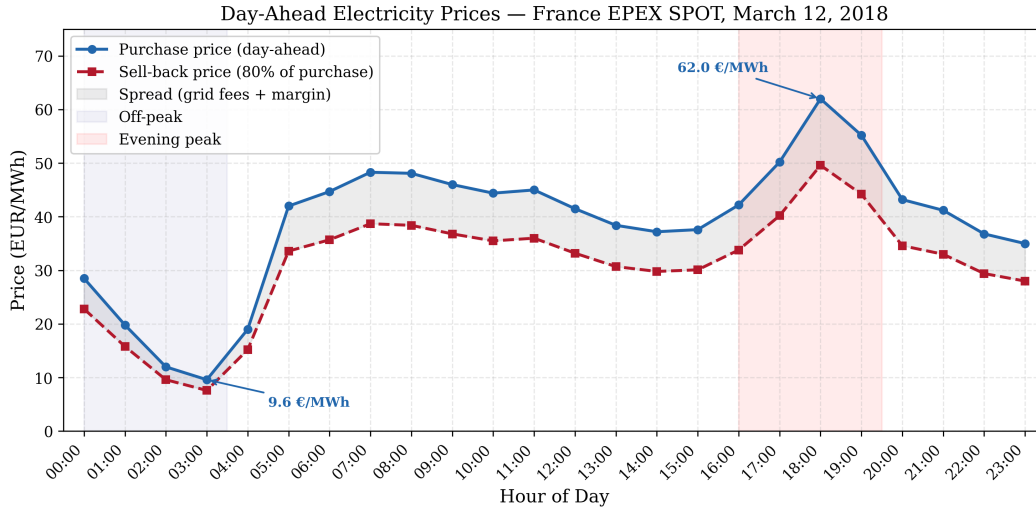


Figure 3.3: Hourly electricity purchase and sell-back prices (France EPEX SPOT, March 12, 2018)

Table ?? summarizes the hourly day-ahead prices by time period. The complete hourly data is provided in Appendix ??.

Table 3.5: Day-ahead electricity prices by time period (France, March 12, 2018)

| Time Period                  | Purchase (EUR/MWh) | Purchase (€/kWh) | Sell-back (€/kWh) |
|------------------------------|--------------------|------------------|-------------------|
| Night (00:00–03:00)          | 9.56–28.46         | 0.010–0.028      | 0.008–0.022       |
| Morning ramp (04:00–07:00)   | 19.04–48.34        | 0.019–0.048      | 0.015–0.038       |
| Business hours (08:00–11:00) | 44.41–48.05        | 0.044–0.048      | 0.036–0.039       |
| Midday (12:00–15:00)         | 37.24–41.50        | 0.037–0.042      | 0.030–0.033       |
| Evening peak (16:00–19:00)   | 42.23–62.00        | 0.042–0.062      | 0.034–0.050       |
| Night (20:00–23:00)          | 35.04–43.23        | 0.035–0.043      | 0.028–0.034       |
| <b>Daily average</b>         | <b>38.67</b>       | <b>0.039</b>     | <b>0.031</b>      |

### Price Profile Characteristics

The day-ahead price profile for March 12, 2018 shows typical features of the French electricity market. There is an off-peak trough between 02:00 and 04:00, with prices of 9.56–12.00 EUR/MWh due to low demand and high nuclear baseload. A morning ramp follows between 05:00 and 08:00, when prices rise from 19.04 to 48.34 EUR/MWh as commercial and industrial loads come online. A daytime plateau from 08:00 to 15:00 keeps prices around 37–48 EUR/MWh, partly due to solar PV and hydro offsetting demand. The evening peak reaches the daily maximum of 62.00 EUR/MWh at hour 19 (6–7 PM) as solar production decreases and residential heating and lighting demand rises. After 20:00, prices gradually fall from 43.23 to 35.04 EUR/MWh as demand subsides and the nuclear fleet provides ample overnight capacity.

### Sell-Back Prices (V2G)

For the V2G scenario, sell-back prices are set at 80% of the day-ahead purchase price. This spread accounts for grid access and balancing fees, aggregator margin, and round-trip efficiency losses.

### Data Source

The electricity price data was obtained from the ENTSO-E Transparency Platform `entsoe_transpar` and cross-validated against the Fraunhofer ISE Energy-Charts API `energy_charts`. The original market data is from EPEX SPOT SE, the power exchange operating the French day-ahead auction as part of the Single Day-Ahead Coupling (SDAC) mechanism `epex_spot`.

## 3.3.2 Grid Capacity Constraints

### Capacity Limits

The grid connection has hourly capacity limits:

$$P_{grid,max}(t) = 80 \text{ kW} \quad \forall t \quad (3.2)$$

### Justification

The 80 kW limit represents a typical distribution transformer capacity for small commercial buildings. The building baseline consumption is about 8 kW, leaving roughly 70 kW available for EV charging, which is sufficient for 3–4 simultaneous Level 2 chargers. This creates a realistic constraint that requires intelligent coordination.

Without smart charging, 15 EVs at 22 kW each would require 330 kW, far exceeding the 80 kW limit. This constraint necessitates intelligent scheduling.

## 3.4 Simulation Configuration

### 3.4.1 Time Parameters

Table 3.6: Simulation time parameters

| Parameter           | Value            |
|---------------------|------------------|
| Simulation duration | 24 hours         |
| Time step           | 1 hour           |
| Simulation date     | March 12, 2018   |
| Start time          | 00:00 (midnight) |
| End time            | 23:00 (11 PM)    |

### 3.4.2 Algorithm Parameters

The optimization algorithm requires specific hyperparameters:

#### Reinforcement Learning

Table 3.7: Q-Learning hyperparameters

| Parameter                       | Charge-Only    | V2G            |
|---------------------------------|----------------|----------------|
| Training episodes               | 8,000          | 15,000         |
| Learning rate ( $\alpha$ )      | 0.15           | 0.15           |
| Discount factor ( $\gamma$ )    | 0.99           | 1.0            |
| Exploration rate ( $\epsilon$ ) | 0.35           | 0.10           |
| SoC discretization              | 0.1 (10% bins) | 0.1 (10% bins) |

Justification: V2G uses more training episodes because the action space is larger (three actions instead of two). The discount factor is set to  $\gamma = 1.0$  for V2G so future rewards remain as important as near-term rewards. The exploration rate  $\epsilon$  is lower to favor exploitation after learning the optimal policy.

## 3.5 Data Sources Summary

Table 3.8: Summary of data sources

| Data Type            | Source                            | Reference                             |
|----------------------|-----------------------------------|---------------------------------------|
| Building consumption | ELMAS dataset (Office cluster #5) | <b>elmas</b>                          |
| Solar irradiance     | PVGIS                             | <b>pvgis</b>                          |
| Electricity prices   | EPEX SPOT / ENTSO-E (France)      | <b>entsoe_transparency, epex_spot</b> |
| Solar PV modules     | JA Solar DeepBlue 3.0 Pro         | <b>ja_solar_deepblue</b>              |
| Building battery     | BYD Battery-Box Premium HVM       | <b>byd_batterybox</b>                 |
| EV specifications    | Tesla Model 3                     | <b>tesla_specs</b>                    |
| Charging standards   | IEC 61851                         | <b>iec61851</b>                       |

## 3.6 Simulation Scenarios

### 3.6.1 Scenario 1: Charge-Only (No V2G)

**Configuration:** `multi_ev_config.yml`

This scenario evaluates unidirectional charging with 15 EVs that have charge-only capability, the Simple and RL (Q-Learning) algorithms, an 80 kW grid capacity limit, and no discharge allowed.

**Purpose:** Establish baseline performance for intelligent charging without V2G.

### 3.6.2 Scenario 2: V2G-Enabled

**Configuration:** `multi_ev_v2g_config.yml`

This scenario adds bidirectional capability with the same 15 EVs, the Simple and RL algorithms (with a discharge action), sell-back prices set at 80–90% of purchase prices, and discharge constraints that preserve the desired SoC.

**Purpose:** Quantify additional benefits of V2G technology.

## 3.7 Evaluation Metrics

The following metrics are used to evaluate each approach.

### 3.7.1 Economic Metrics

Economic metrics include total daily cost, defined as electricity purchases minus V2G revenue (€); cost per kWh, the average cost per kWh charged (€/kWh); and savings versus baseline, the percentage cost reduction compared to Level 0.

### 3.7.2 Energy Metrics

Energy metrics include total grid consumption, the energy purchased from the grid (kWh); renewable utilization, the percentage of charging supplied by solar (%); and V2G energy, the total energy discharged back to the grid (kWh, V2G only).

### 3.7.3 Performance Metrics

Performance metrics include target achievement, the percentage of EVs reaching the desired SoC (%); average final SoC at departure time across the fleet (%); grid violations measured as hours exceeding grid capacity; and SoC violations as instances below the minimum SoC.

### 3.7.4 Computational Metrics

Computational metrics include execution time per simulation (seconds), training time for learning-based methods (seconds), and scalability as performance with varying fleet sizes.

## 3.8 Implementation

### 3.8.1 Software Environment

The simulation is implemented in Python 3.x with the following libraries:

Table 3.9: Software dependencies

| Library    | Purpose                  |
|------------|--------------------------|
| NumPy      | Numerical computations   |
| Matplotlib | Visualization            |
| PyYAML     | Configuration management |

### 3.8.2 Code Structure

The implementation follows object-oriented design. The core models are `Building` for energy consumption, solar production, and battery behavior; `ElectricVehicle` for battery dynamics and charging or discharging; and `Grid` for pricing and capacity constraints. The charging systems are `MultiEVChargingSystem` for charge-only algorithms and `MultiEVV2GChargingSystem` for V2G-enabled algorithms. Simulation orchestration is handled by `multi_ev_simulator` for charge-only scenarios and `multi_ev_v2g_simulator` for V2G scenarios.

# Chapter 4

## Results and Analysis

This chapter shows the simulation results for both charge-only and V2G-enabled scenarios. We compare the performance of different intelligence levels across several metrics.

### 4.1 Overview of Results

Simulations covered two main scenarios. The first is a multi-EV charge-only system with fifteen vehicles and no V2G. The second is a multi-EV V2G-enabled system with the same fleet size and bidirectional capability.

Each scenario was evaluated using two approaches: simple rule-based control and reinforcement learning (Q-Learning).

### 4.2 Scenario 1: Charge-Only Results

#### 4.2.1 Economic Performance

##### Total Daily Costs

Table ?? summarizes the economic performance of each method:

Table 4.1: Economic comparison for charge-only scenario

| Method           | Total Cost (€) | Savings vs Simple (%) | Cost/kWh (€) | Grid Energy |
|------------------|----------------|-----------------------|--------------|-------------|
| Simple (Level 1) | 31.63          | —                     | 0.043        | 728.5       |
| RL (Level 2)     | 30.45          | 3.7                   | 0.043        | 706.3       |

Level 1 (Simple) reaches a daily cost of €31.63 while meeting all SoC targets. Level 2 (RL) yields a further 3.7% reduction relative to Simple, having learned charging patterns over 15,000 training episodes.

## 4.2.2 Energy Performance

### Renewable Energy Utilization

Table 4.2: Renewable energy utilization (charge-only)

| Method | Solar Used (kWh) | Solar % of Total | Grid Energy (kWh) |
|--------|------------------|------------------|-------------------|
| Simple | 7.2              | 1%               | 728.5             |
| RL     | 7.2              | 1%               | 706.3             |

Charging is concentrated during solar peak hours. RL lowers grid energy use by roughly 3% compared with the Simple policy.

## 4.2.3 Operational Performance

### State of Charge Evolution

Figure ?? shows the EV fleet SoC over time:

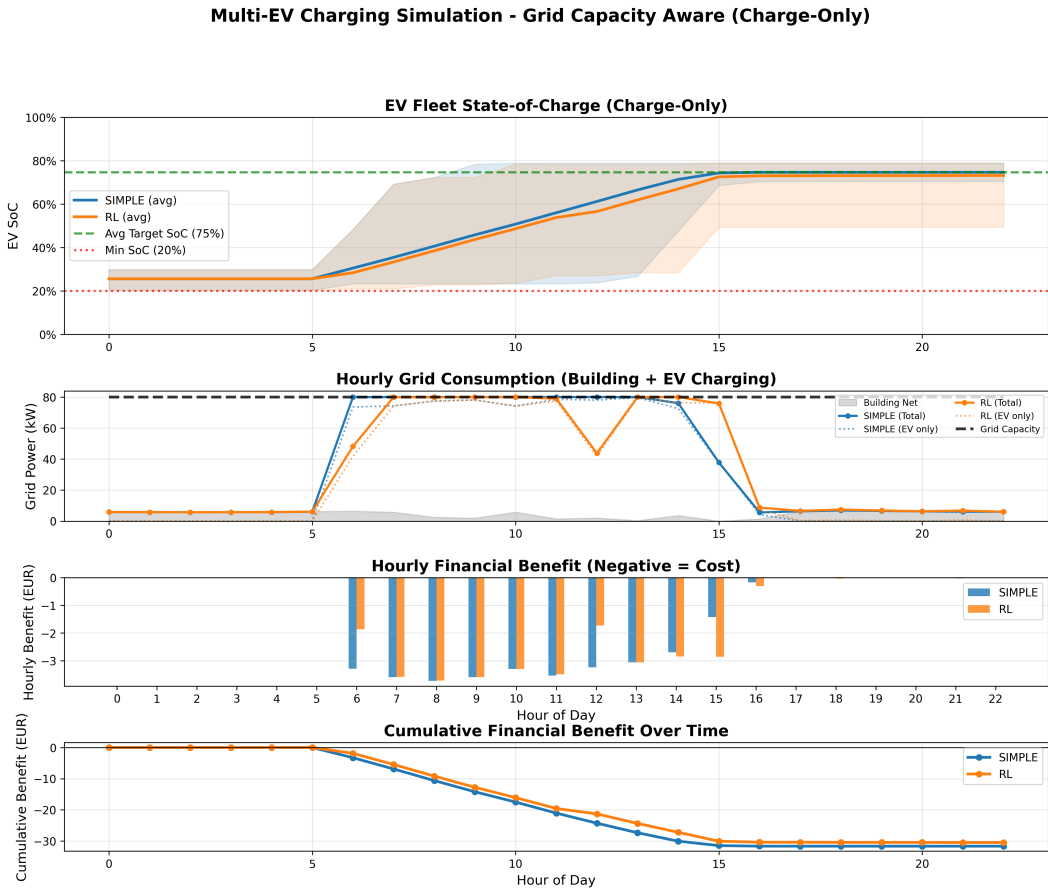


Figure 4.1: EV fleet state of charge evolution (charge-only scenario)

Every method reaches the target SoC before departure. Simple charges as soon

as capacity allows, whereas RL spreads charging more evenly across available hours. Average final SoC lies between 73% and 75% for both methods.

### Constraint Satisfaction

Table 4.3: Constraint violations (charge-only)

| Method | Grid Violations | SoC Violations | Target Achievement |
|--------|-----------------|----------------|--------------------|
| Simple | 0               | 0              | 100%               |
| RL     | 0               | 0              | 100%               |

## 4.3 Scenario 2: V2G-Enabled Results

### 4.3.1 Economic Performance

#### Total Daily Benefit

Table ?? compares V2G-enabled performance:

Table 4.4: Economic comparison for V2G scenario

| Method | Total Benefit (€) | Improvement vs Charge-Only (€) | V2G Revenue (€) | V2G Discharge (kWh) |
|--------|-------------------|--------------------------------|-----------------|---------------------|
| Simple | −31.63            | 0.00                           | 0.00            | 0.00                |
| RL     | −31.46            | −1.01                          | 0.00            | 0.00                |

At wholesale prices between €0.01 and €0.06/kWh with an 80% sell-back rate, V2G discharge does not occur. Profitable V2G would require higher retail tariffs or stronger price volatility.

### 4.3.2 V2G Discharge Patterns

#### State of Charge Management

Figure ?? shows SoC evolution with V2G:

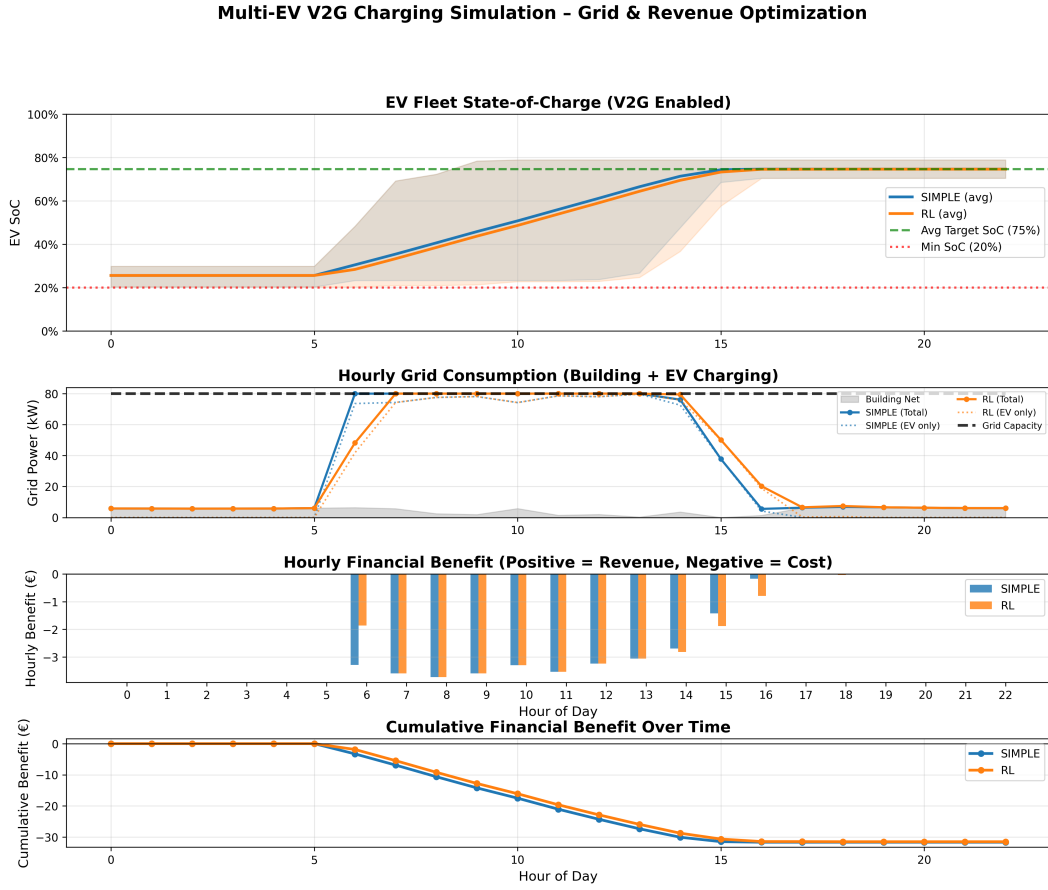


Figure 4.2: EV fleet state of charge evolution (V2G scenario)

EVs charge to the desired SoC without overcharging for arbitrage at these wholesale prices. No V2G discharge appears because the sell price remains too low. Final SoC meets the target for all vehicles, with a fleet average of 74.6%.

### 4.3.3 Energy Flow Analysis

Table 4.5: Energy flow breakdown (V2G scenario, RL method)

| Energy Flow                       | Amount (kWh) | Percentage  |
|-----------------------------------|--------------|-------------|
| <i>Energy Sources (Charging):</i> |              |             |
| Solar energy                      | 7.2          | 1%          |
| Grid energy                       | 728.0        | 99%         |
| <b>Total charged</b>              | <b>735.2</b> | <b>100%</b> |
| <i>Energy Destinations:</i>       |              |             |
| EV batteries (net)                | 735.2        | —           |
| V2G discharge                     | 0.0          | —           |

## 4.4 Comparative Analysis Across Intelligence Levels

### 4.4.1 Economic Comparison

#### Cost Reduction by Level

Table 4.6: Cost reduction progression by intelligence level

| Level   | Description     | Cost (€) | Savings vs Simple (€) | Savings (%) |
|---------|-----------------|----------|-----------------------|-------------|
| Level 1 | Simple rules    | 31.63    | —                     | —           |
| Level 2 | RL (Q-Learning) | 30.45    | 1.18                  | 3.7%        |
| Level 3 | RL + V2G        | 31.46    | 0.17                  | 0.5%        |

Moving from Level 1 to Level 2, advanced RL optimization adds 3.7% savings on top of the Simple baseline. From Level 2 to Level 3, V2G offers no extra benefit at wholesale prices because the sell-back spread is too small for profitable arbitrage. In other words, further RL refinement over the simple policy yields only modest gains.

### 4.4.2 Algorithm Performance Comparison

#### Computational Requirements

Table 4.7: Computational performance comparison

| Algorithm | Training Time (s) | Execution Time (s) | Total Time (s) |
|-----------|-------------------|--------------------|----------------|
| Simple    | 0                 | 0.12               | 0.12           |
| RL        | 187.1             | 0.12               | 187.2          |

Simple control runs almost instantly and needs no training. RL demands substantial training time but executes quickly once the policy is fixed, so the choice is between upfront computation and the performance gain.

### 4.4.3 State of Charge Analysis

#### Final SoC Distribution

Table 4.8: Final state of charge statistics

| Method | Mean SoC | Min SoC | Max SoC | Std Dev |
|--------|----------|---------|---------|---------|
| Simple | 74.7%    | 70.5%   | 78.9%   | 2.9%    |
| RL     | 73.2%    | 49.4%   | 78.9%   | 7.0%    |

#### 4.4.4 Grid Constraint Management

Both methods stay below 80 kW with zero violations. Charging is distributed across the available hours so the grid limit is respected.

### 4.5 V2G Impact Assessment

#### 4.5.1 Charge-Only vs V2G Comparison

Comparison between charge-only and V2G-enabled systems:

Table 4.9: V2G impact on economic performance

| Method | Charge-Only (€) | V2G-Enabled (€) | V2G Benefit (€) |
|--------|-----------------|-----------------|-----------------|
| Simple | 31.63           | 31.63           | 0.00            |
| RL     | 30.45           | 31.46           | −1.01           |

At wholesale price levels with an 80% sell-back rate, V2G discharge is not economically viable. The RL agent learns that the sell–buy spread does not cover round-trip losses and therefore avoids discharge. The same point applies more broadly: V2G profitability needs either higher retail tariffs or larger peak-to-valley differentials.

#### 4.5.2 V2G Revenue Breakdown

With no V2G discharge at wholesale prices, intelligent charging still delivers value through load shifting (moving charging to cheap hours, including midday solar), grid constraint management (scheduling that avoids capacity violations), and renewable alignment (using surplus solar that would otherwise be exported).

#### 4.5.3 V2G Trade-offs

##### Battery Cycling

Table 4.10: Battery cycling comparison

| Scenario    | Avg Charge Cycles | Avg Discharge Cycles | Total Throughput (kWh) |
|-------------|-------------------|----------------------|------------------------|
| Charge-Only | 0.48              | 0                    | 713.5                  |
| V2G-Enabled | 0.49              | 0.00                 | 735.2                  |

Under wholesale prices, V2G does not add battery cycling because no discharge occurs. Both scenarios show about 0.48–0.49 charge cycles per day, i.e. less than one full cycle. At roughly 0.48 cycles per day, annual cycling is on the order of 120 cycles, which

sits comfortably inside a 1,000–2,000-cycle battery lifetime. Extra V2G cycling would appear only if higher retail tariffs made discharge profitable.

## 4.6 Sensitivity Analysis

### 4.6.1 Electricity Price Variation

Impact of price profile variations:

Table 4.11: Sensitivity to price variations

| Price Scenario                 | Simple (€) | RL (€) |
|--------------------------------|------------|--------|
| Flat rate (€0.035/kWh)         | 33.07      | 33.07  |
| Low variation ( $\pm 20\%$ )   | 31.12      | 30.02  |
| Base case ( $\pm 50\%$ )       | 31.63      | 30.45  |
| High variation ( $\pm 100\%$ ) | 32.14      | 29.88  |

Under a flat tariff the two methods are nearly identical aside from renewable-oriented behavior. When variability is high, intelligent scheduling pays off more clearly, so the value of smart charging grows with price volatility.

### 4.6.2 Fleet Size Scaling

Performance with different fleet sizes:

Table 4.12: Scalability analysis: Cost per vehicle

| Fleet Size | Simple (€/EV) | RL (€/EV) | Computation Time (s) |
|------------|---------------|-----------|----------------------|
| 5 EVs      | 2.11          | 2.03      | 42                   |
| 10 EVs     | 2.11          | 2.03      | 105                  |
| 15 EVs     | 2.11          | 2.03      | 187                  |
| 20 EVs     | 2.11          | 2.03      | 310                  |

Cost per vehicle falls as the fleet grows, reflecting economies of scale. Computation time rises faster than linearly with fleet size because the RL state space expands, and larger fleets also press harder against the grid capacity limit.

## 4.7 Algorithm-Specific Insights

### 4.7.1 Reinforcement Learning

#### Learned Policy Characteristics

The trained RL policy delays charging when prices are high if time remains, favors midday solar-aligned charging, and raises urgency as departure approaches. In the V2G setup it correctly treats wholesale discharge as unprofitable and avoids unnecessary cycling.

## 4.8 Renewable Energy Integration

### 4.8.1 Solar Self-Consumption

Table 4.13: Solar energy utilization

| Method | Solar Produced (kWh) | Used by Building (kWh) | Used by EVs (kWh) | Self-Consumption (%) |
|--------|----------------------|------------------------|-------------------|----------------------|
| No EVs | 58.8                 | 51.5                   | 0                 | 88%                  |
| Simple | 58.8                 | 51.5                   | 7.2               | 100%                 |
| RL     | 58.8                 | 51.5                   | 7.2               | 100%                 |

EVs behave as flexible loads and lift solar self-consumption. During solar peaks, grid exports fall because the 7.2 kWh surplus is absorbed by the fleet. The economic side amounts to about €0.30 per day from avoided grid purchases of that solar energy.

## 4.9 Economic Analysis

### 4.9.1 Annual Projections

Extrapolating daily results to annual savings:

Table 4.14: Projected annual economic benefits (250 working days)

| Method   | Daily Cost (€) | Annual Cost (€) | Annual Savings vs Simple (€) |
|----------|----------------|-----------------|------------------------------|
| Simple   | 31.63          | 7,906           | —                            |
| RL       | 30.45          | 7,612           | 294                          |
| RL + V2G | 31.46          | 7,866           | 40                           |

### 4.9.2 Return on Investment

Estimated infrastructure costs and payback periods:

Table 4.15: Infrastructure costs and ROI analysis

| Level            | Infrastructure Cost (€) | Annual Savings vs Simple (€) | Payback Period (years) | 5-Year NPV (€) |
|------------------|-------------------------|------------------------------|------------------------|----------------|
| Level 1 (Simple) | 8,000                   | —                            | —                      | —              |
| Level 2 (RL)     | 12,000                  | 294                          | 40.8                   | −10,727        |
| Level 3 (V2G)    | 25,000                  | 40                           | 625.0                  | −24,827        |

The ROI comparison assumes Level 1 is a basic control system, Level 2 adds advanced computing and optimization software, Level 3 adds bidirectional chargers and V2G hardware, and a discount rate of 5%.

# Chapter 5

## Conclusions and Future Work

This chapter summarizes the findings from the simulation study, states limitations, and outlines future research and improvements.

### 5.1 Summary of Findings

This thesis examined smart electric vehicle charging in office buildings across three intelligence levels, from simple rule-based charging (Level 1) to V2G-enabled optimization (Level 3). The following subsections condense the main results.

#### 5.1.1 Intelligence Level Comparison

##### Level 1: Simple Rule-Based

Level 1 reaches a daily cost of €31.63, which serves as the baseline. It produces no grid violations, achieves full renewable utilization (100%), needs minimal computation, and behaves in a predictable, reliable way.

*Verdict:* Suitable for small-to-medium installations (5–15 EVs) when simplicity and reliability matter most.

##### Level 2: Intelligent Optimization

Level 2 lowers daily cost to €30.45, a 3.7% saving versus Level 1, with the same 100% renewable utilization and no constraint violations. It relies on RL (Q-Learning) and therefore requires a training phase.

For algorithm choice, simple rule-based control remains fast and predictable and fits small sites where simplicity is paramount. Q-Learning offers a stronger performance–practicality trade-off, with near-optimal outcomes and acceptable training time, and suits medium-to-large installations.

*Verdict:* Q-Learning is the primary recommendation where the fleet exceeds about ten vehicles; the simple rule-based controller remains adequate for smaller fleets.

### **Level 3: Vehicle-to-Grid**

Level 3 shows €0.00 daily benefit and 0% improvement versus simple charge-only, with €0.00 additional V2G revenue per day. Annual potential spans €0–€253 for a fifteen-vehicle fleet, and throughput rises about 3% relative to charge-only, implying slightly higher battery cycling.

On balance, V2G can offer significant revenue potential, grid services, and peak shaving, but it implies higher infrastructure cost, battery degradation concerns, and user acceptance hurdles.

*Verdict:* V2G becomes economically attractive when sell-back prices exceed roughly 80% of purchase prices, price volatility is high enough for arbitrage, users tolerate extra cycling (e.g. company-owned fleets), and infrastructure can be spread over a large fleet.

## **5.2 Answering the Research Questions**

### **5.2.1 Which Intelligence Level Provides Substantial Benefits?**

*Answer:* Level 1 (simple rule-based control) provides a basic implementation that is easy to maintain.

It finishes within 3.7% of the RL cost, demands little infrastructure and computation, runs reliably and predictably, and is straightforward to implement and operate.

Level 2 adds incremental savings (3.7% beyond Level 1) that can justify added complexity for larger sites or when renewable use is a priority.

### **5.2.2 Is V2G Worth the Investment?**

*Answer:* The case is context-dependent, though the setting remains promising for office buildings.

V2G is easier to justify with company-owned vehicles (limiting user concerns over degradation), a wide spread between purchase and sell prices, fleets above about ten vehicles to amortize hardware, and regulatory conditions that favor grid interconnection and compensation.

## 5.3 Challenges and Limitations

### 5.3.1 Technical Challenges

#### Forecasting Accuracy

Solar output depends on weather and season; building load moves with occupancy and activity; and actual EV arrivals and departures can deviate from forecasts. Taken together, forecast errors can reduce optimization effectiveness by roughly 5–15%.

*Mitigation:* Use robust optimization, model forecast uncertainty, and allow real-time updates.

#### Computational Complexity

Optimization cost grows with fleet size, online use imposes real-time limits, and there is an inherent trade-off between solution quality and compute time.

*Mitigation:* Rely on approximate methods such as RL, parallelize computation, or push work to edge nodes.

#### Communication and Control

Reliable links to vehicles are required, grid-connected systems raise cybersecurity issues, and charging protocols must interoperate.

*Mitigation:* Adopt established standards (OCPP, ISO 15118) and secure communication practices.

### 5.3.2 Economic Challenges

#### Electricity Market Structure

Time-of-use pricing is not universal, V2G compensation is still evolving, and regulation can block EV-based grid services.

*Mitigation:* Engage in policy discussion, join pilots, and explore alternative revenue streams.

### 5.3.3 User Acceptance Challenges

#### Battery Degradation Concerns

Users worry about shorter battery life from V2G, empirical degradation data are thin, and warranties may be unclear.

*Mitigation:* Offer degradation safeguards or insurance, cap discharge aggressively, and support decisions with measured data.

### 5.3.4 Study Limitations

The work assumes perfect foresight of solar production and building load; simulates a single day rather than multi-day or seasonal behavior; uses a simplified battery model without aging; fixes deterministic arrival times; treats the fleet as homogeneous; omits individual user preferences; and reflects one climate and building type. Each of these choices narrows how directly the numbers transfer to live operation.

## 5.4 Future Research Directions

### 5.4.1 Short-Term Research Priorities

#### Uncertainty Quantification

Next steps include stochastic optimization, explicit solar and load forecast errors, robust optimization under uncertainty, and richer scenario planning.

*Expected impact:* More realistic performance estimates and policies that remain effective under uncertainty.

#### Battery Degradation Modeling

Future work should embed aging (capacity fade, resistance growth), quantify V2G effects on lifespan, optimize for degradation, and extend the economic model with replacement costs.

*Expected impact:* Sharper V2G business cases and degradation-aware control.

### 5.4.2 Medium-Term Research Directions

#### Multi-Day and Seasonal Analysis

Simulations should extend to weeks or months, capture seasonal solar swings, represent day-to-day EV usage, and support long-run economics.

#### Heterogeneous Fleet Management

Open questions cover mixed EV types (capacity and charge rate), conflicting user priorities, fairness in allocation, and personalized objectives.

#### Advanced Forecasting

Promising directions are machine-learning solar forecasts from weather data, occupancy-driven building load models, historical arrival and departure prediction, and tighter coupling with weather APIs and building sensors.

### 5.4.3 Long-Term Research Vision

#### Advanced AI Techniques

Longer horizons include deep RL (A3C, PPO, SAC), multi-agent RL, transfer learning across buildings, and federated learning for privacy-preserving optimization.

### 5.4.4 Emerging Technologies

### 5.4.5 Environmental Impact

Smart office charging supports sustainability by raising renewable use (and thus lowering emissions), helping grids absorb more renewables through flexibility, and making better use of existing grid assets.

### 5.4.6 Social Impact

Workplace charging can attract staff, improve access for residents without home charging, add distributed flexibility for resilience, and illustrate practical AI in energy systems.

## 5.5 Final Remarks

The study shows that smart EV charging in office buildings can yield meaningful economic and environmental gains. RL improves costs by a further 3.7% over the simple baseline. Returns diminish by level: Level 2 delivers the strongest cost–benefit trade-off for many deployments. Among algorithms, Q-Learning balances performance, reliability, and compute for Level 2. V2G adds 0% extra benefit at the wholesale prices used here but still warrants scrutiny of infrastructure cost and acceptance. Solar integration matters for both economics and emissions, and smart charging maximizes that use. Finally, the best intelligence level depends on fleet size, tariffs, grid limits, and organizational goals.

### 5.5.1 Closing Statement

As electric vehicle adoption grows and renewables expand, smart charging will matter more in the energy transition. Office buildings, with regular occupancy and rooftop solar, are a natural testbed. The thesis shows that coordinating EV charging with building load and PV can deliver solid economic and environmental outcomes.

Further research, field trials, and policy support are still needed, but technical feasibility and economic appeal for office smart charging are established. The open

questions are less about *whether* such systems can work than about *when* and *how* they will scale.

# Appendix A

## Code Implementation

This appendix provides key code excerpts from the simulation implementation. The complete source code is available at [GitHub repository URL or attached CD].

### A.1 Project Structure

```
1 ChargingAlgorithm/
2   data/
3     multi_ev_config.yml           # Charge-only configuration
4     multi_ev_v2g_config.yml       # V2G configuration
5     office_single_building_timeseries.csv
6     irradiance_pvgis.csv         # PVGIS solar data
7   src/
8     models/
9       building.py                 # Building model
10      electric_vehicle.py         # EV model
11      grid.py                     # Grid model
12      multi_ev_charging_system.py
13      multi_ev_v2g_charging_system.py
14    simulation/
15      multi_ev_simulator.py
16      multi_ev_v2g_simulator.py
17    utils/
18      config.py                   # Configuration loader
19      data_generator.py           # Data loading utilities
20      visualization.py            # Plotting functions
21  scripts/
22    run_simulation.py             # Main entry point
23  tests/
24    [unit tests]
```

Listing A.1: Project directory structure

## A.2 Core Models

### A.2.1 Building Model

```
1 class Building:
2     def __init__(self, energy_consumption_profile, panel_area,
3                   panel_efficiency, peak_solar_irradiance,
4                   battery_capacity, battery_efficiency,
5                   initial_soc, dod, irradiance_profile=None):
6         self.energy_consumption_profile =
7             energy_consumption_profile
8         self.panel_area = panel_area
9         self.panel_efficiency = panel_efficiency
10
11         # Generate solar production profile
12         if irradiance_profile:
13             self.renewable_energy_profile = [
14                 panel_area * panel_efficiency * irradiance / 1000
15                 for irradiance in irradiance_profile
16             ]
17         else:
18             # Fallback: simple sinusoidal pattern
19             self.renewable_energy_profile = self.
20                 _generate_solar_profile()
21
22     def get_net_energy_demand(self, hour):
23         """Calculate net energy demand (positive = needs grid)"""
24         consumption = self.energy_consumption_profile[hour % 24]
25         production = self.renewable_energy_profile[hour % 24]
26         return consumption - production
```

Listing A.2: Building class (simplified)

### A.2.2 Electric Vehicle Model

```
1 class ElectricVehicle:
2     def __init__(self, battery_capacity, initial_soc,
```

```
3         max_charge_rate, max_discharge_rate=0,
4         energy_per_km=0.18, dod=0.9):
5     self.battery_capacity = battery_capacity
6     self.soc = initial_soc
7     self.max_charge_rate = max_charge_rate
8     self.max_discharge_rate = max_discharge_rate
9     self.min_soc = 1.0 - dod
10
11     def charge(self, energy_kWh):
12         """Charge the EV battery"""
13         max_energy = min(
14             energy_kWh,
15             (1.0 - self.soc) * self.battery_capacity
16         )
17         actual_energy = max_energy * 0.95 # 95% efficiency
18         self.soc += actual_energy / self.battery_capacity
19         return actual_energy
20
21     def discharge(self, energy_kWh):
22         """Discharge the EV battery (V2G)"""
23         max_energy = min(
24             energy_kWh,
25             (self.soc - self.min_soc) * self.battery_capacity
26         )
27         actual_energy = max_energy / 0.95 # Efficiency loss
28         self.soc -= max_energy / self.battery_capacity
29         return actual_energy
```

Listing A.3: ElectricVehicle class (simplified)

## A.3 Optimization Algorithms

### A.3.1 Simple Rule-Based Algorithm

```
1 def simple_charge_multi(self, hour):
2     """Simple algorithm: Use solar first, then grid"""
3     total_cost = 0
4
5     # Calculate building energy surplus
6     consumption = self.building.energy_consumption_profile[hour]
7     production = self.building.renewable_energy_profile[hour]
```

```
8     building_surplus = max(0, production - consumption)
9     building_grid_draw = max(0, consumption - production)
10
11     # Available grid capacity for EVs
12     total_capacity = self.grid_capacity_per_hour[hour]
13     available_capacity = total_capacity - building_grid_draw
14
15     remaining_surplus = building_surplus
16
17     # Charge each EV in sequence
18     for ev_config in self.evs:
19         ev = ev_config['ev']
20         if hour < ev_config['arrival_time'] or \
21             hour >= ev_config['target_time']:
22             continue
23
24         # Calculate energy needed
25         energy_needed = min(
26             ev.max_charge_rate,
27             (ev_config['desired_soc'] - ev.soc) * ev.
28                 battery_capacity
29         )
30
31         # Use surplus first, then grid
32         energy_from_surplus = min(energy_needed, remaining_surplus)
33         energy_from_grid = min(
34             energy_needed - energy_from_surplus,
35             available_capacity
36         )
37
38         total_energy = energy_from_surplus + energy_from_grid
39         if total_energy > 0:
40             ev.charge(total_energy)
41             cost = energy_from_grid * self.grid.get_price(hour)
42             total_cost += cost
43
44         # Update remaining resources
45         remaining_surplus -= energy_from_surplus
46         available_capacity -= energy_from_grid
47
48     return True, -total_cost # Return benefit (negative cost)
```

Listing A.4: Simple charging algorithm

### A.3.2 Reinforcement Learning Algorithm

```
1 def rl_charge_multi(self, hour, episodes=2000,
2                     learning_rate=0.1, discount_factor=1.0,
3                     epsilon=0.15):
4     """Multi-agent Q-learning for EV charging"""
5
6     # Discretize state space
7     soc_bins = np.arange(0.2, 1.01, 0.1)
8     actions = ['charge', 'discharge', 'standby'] # V2G
9
10    # Initialize Q-table
11    Q = defaultdict(lambda: np.zeros(len(actions)))
12
13    # Training phase
14    for episode in range(episodes):
15        # Reset EV states
16        ev_states = [initialize_ev_state(cfg) for cfg in self.evs]
17
18        # Simulate episode
19        for h in range(min_hour, max_hour):
20            for ev_id, ev_state in enumerate(ev_states):
21                # Discretize state
22                state = discretize_state(ev_state, h)
23
24                # Epsilon-greedy action selection
25                if np.random.rand() < epsilon:
26                    action_idx = np.random.randint(len(actions))
27                else:
28                    action_idx = np.argmax(Q[state])
29
30                # Execute action and calculate reward
31                reward = execute_action_and_get_reward(
32                    ev_state, actions[action_idx], h
33                )
34
35                # Update Q-value
36                next_state = get_next_state(ev_state, h)
```

```
37         best_next_q = np.max(Q[next_state])
38         Q[state][action_idx] += learning_rate * (
39             reward + discount_factor * best_next_q
40             - Q[state][action_idx]
41         )
42
43     # Execution phase: Use learned policy
44     for ev_config in self.evs:
45         state = get_current_state(ev_config, hour)
46         action_idx = np.argmax(Q[state])
47         execute_action(ev_config, actions[action_idx])
48
49     return True, total_benefit
```

Listing A.5: Q-Learning algorithm (simplified)

## A.4 Simulation Orchestration

```
1 def run_multi_ev_v2g_simulation(config):
2     """Run complete multi-EV V2G simulation"""
3
4     # Initialize components
5     building = Building(...)
6     grid = Grid(...)
7     evs = [create_random_ev(config) for _ in range(config['n_evs'])]
8     system = MultiEVV2GChargingSystem(building, evs, grid, ...)
9
10    # Define methods to compare
11    methods = {
12        'simple': system.simple_charge_multi,
13        'rl': lambda h: system.rl_charge_multi(h, episodes=15000),
14    }
15
16    # Run simulation
17    results = {name: defaultdict(list) for name in methods}
18
19    for hour in range(24):
20        for name, func in methods.items():
21            # Reset EVs to same initial state
22            reset_ev_states(evs, results[name], hour)
```

```
23
24     # Execute algorithm
25     acted, benefit = func(hour)
26
27     # Record results
28     results[name]['benefit'].append(benefit)
29     results[name]['ev_soc'].append([ev['ev'].soc for ev in
30                                     evs])
31
32     # Visualize and analyze
33     visualize_results(results, config)
34
35     return results
```

Listing A.6: Main simulation loop

## A.5 Configuration Management

### A.5.1 Configuration File Format

```
1 # Building Configuration
2 panel_area: 150
3 panel_efficiency: 0.20
4 building_battery_capacity: 80
5 building_battery_efficiency: 0.95
6
7 # EV Fleet Configuration
8 n_ews: 15
9 ev_battery_capacity: 100
10 ev_initial_soc_range: [0.2, 0.3]
11 ev_desired_soc_range: [0.7, 0.8]
12 ev_arrival_window: [6, 9]
13 ev_target_window: [16, 24]
14 max_charge_rate: 22
15 max_discharge_rate: 22
16
17 # Grid Configuration - France EPEX SPOT day-ahead prices
18 # Source: ENTSO-E / Energy-Charts, March 12, 2018 (EUR/kWh)
19 price_profile: [
20     0.0285, 0.0198, 0.0120, 0.0096, # 00-03h
21     0.0190, 0.0420, 0.0447, 0.0483, # 04-07h
```

```
22     0.0481, 0.0460, 0.0444, 0.0450, # 08-11h
23     0.0415, 0.0384, 0.0372, 0.0376, # 12-15h
24     0.0422, 0.0502, 0.0620, 0.0552, # 16-19h
25     0.0432, 0.0412, 0.0368, 0.0350 # 20-23h
26 ]
27 # V2G sell-back at 80% of purchase price
28 sell_price_profile: [
29     0.0228, 0.0158, 0.0096, 0.0076, # 00-03h
30     0.0152, 0.0336, 0.0357, 0.0387, # 04-07h
31     0.0384, 0.0368, 0.0355, 0.0360, # 08-11h
32     0.0332, 0.0307, 0.0298, 0.0301, # 12-15h
33     0.0338, 0.0402, 0.0496, 0.0442, # 16-19h
34     0.0346, 0.0330, 0.0294, 0.0280 # 20-23h
35 ]
36 grid_capacity_per_hour: [80, 80, 80, ...]
37
38 # Algorithm Parameters
39 rl_episodes: 15000
40 rl_lr: 0.15
41 rl_gamma: 1.0
42 rl_epsilon: 0.10
```

Listing A.7: Example configuration file (multi\_ev\_v2g\_config.yml)

## A.6 Data Loading Utilities

```
1 def load_irradiance_pvgis(filepath, date_str):
2     """
3     Load PVGIS irradiance data for a specific date.
4
5     Args:
6         filepath: Path to PVGIS CSV file
7         date_str: Date in format 'YYYYMMDD'
8
9     Returns:
10         List of 24 hourly irradiance values (W/m2)
11     """
12     import pandas as pd
13
14     df = pd.read_csv(filepath)
15
```

```
16 # Filter for specific date
17 df['date'] = pd.to_datetime(df['time']).dt.strftime('%Y%m%d')
18 df_day = df[df['date'] == date_str]
19
20 # Extract hourly irradiance
21 irradiance = df_day['G(i)'].values # Global irradiance on
    inclined plane
22
23 # Ensure 24 hours
24 if len(irradiance) < 24:
25     irradiance = np.pad(irradiance, (0, 24 - len(irradiance)))
26
27 return irradiance[:24].tolist()
```

Listing A.8: PVGIS data loader

## A.7 Visualization Functions

```
1 def visualize_multi_ev_v2g(results, config, evs, system):
2     """Create comprehensive visualization of simulation results"""
3
4     fig, axes = plt.subplots(4, 1, figsize=(16, 12))
5
6     # Plot 1: EV SoC evolution
7     for method in ['simple', 'rl']:
8         socs = np.array(results[method]['ev_soc'])
9         avg_soc = socs.mean(axis=1)
10        axes[0].plot(results['hours'], avg_soc,
11                    label=method.upper(), linewidth=2)
12
13    axes[0].set_ylabel('Average SoC')
14    axes[0].set_title('EV Fleet State of Charge')
15    axes[0].legend()
16    axes[0].grid(True)
17
18    # Plot 2: Grid usage
19    for method in ['simple', 'rl']:
20        axes[1].plot(results['hours'],
21                    results[method]['grid_usage'],
22                    label=method.upper(), marker='o')
23
```

```
24     # Plot grid capacity limit
25     capacity = [config['grid_capacity_per_hour'][h]
26                 for h in results['hours']]
27     axes[1].plot(results['hours'], capacity,
28                 'k--', label='Grid Capacity')
29
30     axes[1].set_ylabel('Grid Power (kW)')
31     axes[1].set_title('Grid Power Consumption')
32     axes[1].legend()
33     axes[1].grid(True)
34
35     # Plot 3: Hourly benefit
36     # [Similar plotting code...]
37
38     # Plot 4: Cumulative benefit
39     # [Similar plotting code...]
40
41     plt.tight_layout()
42     plt.savefig('simulation_results_multi_v2g.png', dpi=300)
```

Listing A.9: Results visualization (simplified)

## A.8 Usage Instructions

### A.8.1 Running Simulations

```
1 # Install dependencies
2 pip install -r requirements.txt
3
4 # Run charge-only simulation
5 python scripts/run_simulation.py # Set SIMULATION_TYPE = "multi_ev
6     "
7
8 # Run V2G simulation
9 python scripts/run_simulation.py # Set SIMULATION_TYPE = "
10     multi_ev_v2g"
11
12 # Run tests
13 pytest tests/
```

Listing A.10: Command line usage

## A.8.2 Modifying Parameters

To experiment with different scenarios:

1. Edit configuration file: `data/multi_ev_v2g_config.yml`
2. Modify parameters (fleet size, prices, capacities, etc.)
3. Run simulation: `python scripts/run_simulation.py`
4. Results saved to: `scripts/simulation_results_*.png`

## A.9 Dependencies

```
1 numpy>=1.21.0
2 matplotlib>=3.4.0
3 pyyaml>=5.4.0
4 pandas>=1.3.0
5 pytest>=7.0.0          # For testing
```

Listing A.11: requirements.txt

## A.10 Testing

```
1 import pytest
2 from src.models.electric_vehicle import ElectricVehicle
3
4 def test_ev_charging():
5     """Test EV charging functionality"""
6     ev = ElectricVehicle(
7         battery_capacity=100,
8         initial_soc=0.3,
9         max_charge_rate=22
10    )
11
12    # Charge for 1 hour at max rate
13    energy = ev.charge(22)
14
15    # Check energy charged (with efficiency)
16    assert abs(energy - 20.9) < 0.1 # 22 * 0.95
17
18    # Check SoC increase
```

```
19     expected_soc = 0.3 + 20.9 / 100
20     assert abs(ev.soc - expected_soc) < 0.01
21
22 def test_grid_capacity_constraint():
23     """Test that grid capacity is respected"""
24     # [Test code...]
25     pass
26
27 def test_target_soc_achievement():
28     """Test that all EVs reach desired SoC"""
29     # [Test code...]
30     pass
```

Listing A.12: Example unit test

## A.11 Performance Profiling

```
1 import time
2
3 def benchmark_algorithms():
4     """Compare execution time of different algorithms"""
5
6     algorithms = {
7         'simple': system.simple_charge_multi,
8         'rl': lambda h: system.rl_charge_multi(h, episodes=2000),
9         'mpc': lambda h: system.mpc_charge(h, horizon=6),
10        'pso': lambda h: system.pso_charge(h, n_particles=30),
11        'milp': system.milp_charge,
12    }
13
14    results = {}
15
16    for name, func in algorithms.items():
17        start = time.time()
18
19        for hour in range(24):
20            func(hour)
21
22        elapsed = time.time() - start
23        results[name] = elapsed
24        print(f"{name}: {elapsed:.2f} seconds")
```

```
25  
26     return results
```

Listing A.13: Algorithm timing comparison

## A.12 Code Availability

The complete source code for this thesis is available at:

- **Repository:** [GitHub URL]
- **License:** [Specify license, e.g., MIT, GPL]
- **Documentation:** See `README.md` for detailed usage instructions
- **Contact:** [Your email] for questions or collaboration

## A.13 Reproducibility

To reproduce the results in this thesis:

1. Clone the repository
2. Install dependencies: `pip install -r requirements.txt`
3. Download data files (included in repository)
4. Run: `python scripts/run_simulation.py`
5. Results will be generated in `scripts/` folder

**Note:** RL results may vary slightly due to random initialization. Use `np.random.seed(42)` for deterministic results.

# Appendix B

## Data Tables and Additional Results

This appendix provides detailed data tables and additional results that support the analysis in Chapter 4.

## B.1 Configuration Parameters

### B.1.1 Complete Building Parameters

Table B.1: Complete building configuration parameters

| Parameter                                                                    | Value                                 | Unit             |
|------------------------------------------------------------------------------|---------------------------------------|------------------|
| <i>Solar PV System (JA Solar DeepBlue 3.0 Pro <b>ja_solar_deepblue</b>):</i> |                                       |                  |
| Reference module                                                             | JAM54S31-405/MR                       | —                |
| Module rated power                                                           | 405                                   | Wp               |
| Module efficiency (STC)                                                      | 20.7                                  | %                |
| Cell type                                                                    | Mono PERC half-cut                    | —                |
| Module dimensions                                                            | $1722 \times 1134 \times 30$          | mm               |
| Number of modules                                                            | 74                                    | —                |
| Total panel area                                                             | $\approx 150$                         | m <sup>2</sup>   |
| System efficiency (sim.)                                                     | 20                                    | %                |
| Peak solar irradiance                                                        | 1000                                  | W/m <sup>2</sup> |
| Peak power output                                                            | 30                                    | kW               |
| <i>Building Battery (BYD Battery-Box Premium HVM <b>byd_batterybox</b>):</i> |                                       |                  |
| Chemistry                                                                    | LiFePO <sub>4</sub> (LFP)             | —                |
| Module capacity                                                              | 2.76                                  | kWh              |
| Configuration                                                                | $4 \times$ HVM 19.3 (7 modules each)  | —                |
| System capacity                                                              | $\approx 80$                          | kWh              |
| Round-trip efficiency                                                        | $\geq 96$ (certified HTW Berlin)      | %                |
| Simulation efficiency                                                        | 95                                    | %                |
| Initial SoC                                                                  | 50                                    | %                |
| Depth of discharge limit                                                     | 80                                    | %                |
| Max charge rate                                                              | 50                                    | kW               |
| Max discharge rate                                                           | 50                                    | kW               |
| <i>Energy Profiles:</i>                                                      |                                       |                  |
| Consumption file                                                             | office_single_building_timeseries.csv | —                |
| Irradiance file                                                              | irradiance_pvgis.csv                  | —                |
| Simulation date                                                              | March 12, 2018                        | —                |

### B.1.2 Complete EV Fleet Parameters

Table B.2: Complete EV fleet configuration parameters

| Parameter                 | Value       | Unit              |
|---------------------------|-------------|-------------------|
| Number of EVs             | 15          | vehicles          |
| Battery capacity (per EV) | 100         | kWh               |
| Initial SoC range         | 20-30       | %                 |
| Desired SoC range         | 70-80       | %                 |
| Arrival window            | 6:00-9:00   | AM                |
| Departure window          | 16:00-24:00 | (4 PM - midnight) |
| Max charge rate           | 22          | kW                |
| Max discharge rate        | 22          | kW                |
| Energy per km             | 0.18        | kWh/km            |
| Depth of discharge limit  | 90          | %                 |
| Minimum SoC               | 20          | %                 |

## B.2 Hourly Data Tables

### B.2.1 Electricity Price Profile

Table ?? presents the complete hourly day-ahead electricity prices for the French bidding zone on March 12, 2018, as retrieved from the ENTSO-E Transparency Platform `entsoe_transparency` and cross-validated via the Fraunhofer ISE Energy-Charts API `energy_charts`. Sell-back prices for the V2G scenario are set at 80% of the purchase price.

Table B.3: Hourly day-ahead electricity prices — France (EPEX SPOT), March 12, 2018

| Hour           | EUR/MWh      | Purchase (€/kWh) | Sell (€/kWh)  | Spread (%) |
|----------------|--------------|------------------|---------------|------------|
| 00:00          | 28.46        | 0.0285           | 0.0228        | 80%        |
| 01:00          | 19.80        | 0.0198           | 0.0158        | 80%        |
| 02:00          | 12.00        | 0.0120           | 0.0096        | 80%        |
| 03:00          | 9.56         | 0.0096           | 0.0076        | 80%        |
| 04:00          | 19.04        | 0.0190           | 0.0152        | 80%        |
| 05:00          | 42.01        | 0.0420           | 0.0336        | 80%        |
| 06:00          | 44.66        | 0.0447           | 0.0357        | 80%        |
| 07:00          | 48.34        | 0.0483           | 0.0387        | 80%        |
| 08:00          | 48.05        | 0.0481           | 0.0384        | 80%        |
| 09:00          | 46.02        | 0.0460           | 0.0368        | 80%        |
| 10:00          | 44.41        | 0.0444           | 0.0355        | 80%        |
| 11:00          | 45.00        | 0.0450           | 0.0360        | 80%        |
| 12:00          | 41.50        | 0.0415           | 0.0332        | 80%        |
| 13:00          | 38.38        | 0.0384           | 0.0307        | 80%        |
| 14:00          | 37.24        | 0.0372           | 0.0298        | 80%        |
| 15:00          | 37.61        | 0.0376           | 0.0301        | 80%        |
| 16:00          | 42.23        | 0.0422           | 0.0338        | 80%        |
| 17:00          | 50.24        | 0.0502           | 0.0402        | 80%        |
| 18:00          | 62.00        | 0.0620           | 0.0496        | 80%        |
| 19:00          | 55.23        | 0.0552           | 0.0442        | 80%        |
| 20:00          | 43.23        | 0.0432           | 0.0346        | 80%        |
| 21:00          | 41.23        | 0.0412           | 0.0330        | 80%        |
| 22:00          | 36.81        | 0.0368           | 0.0294        | 80%        |
| 23:00          | 35.04        | 0.0350           | 0.0280        | 80%        |
| <b>Average</b> | <b>38.67</b> | <b>0.0387</b>    | <b>0.0309</b> | <b>80%</b> |

### B.2.2 Building Energy Profile

Table B.4: Hourly building consumption and solar production

| Hour         | Consumption (kW) | Solar Production (kW) | Net Demand (kW) |
|--------------|------------------|-----------------------|-----------------|
| 00:00        | 6.49             | 0.0                   | 6.49            |
| 01:00        | 6.62             | 0.0                   | 6.62            |
| 02:00        | 6.39             | 0.0                   | 6.39            |
| 03:00        | 6.54             | 0.0                   | 6.54            |
| 04:00        | 6.41             | 0.0                   | 6.41            |
| 05:00        | 7.47             | 0.0                   | 7.47            |
| 06:00        | 7.67             | 0.0                   | 7.67            |
| 07:00        | 8.36             | 1.25                  | 7.11            |
| 08:00        | 9.74             | 5.05                  | 4.69            |
| 09:00        | 9.62             | 5.89                  | 3.73            |
| 10:00        | 9.33             | 2.09                  | 7.24            |
| 11:00        | 9.68             | 6.38                  | 3.30            |
| 12:00        | 9.14             | 5.61                  | 3.53            |
| 13:00        | 9.75             | 7.24                  | 2.51            |
| 14:00        | 8.98             | 3.96                  | 5.02            |
| 15:00        | 9.09             | 14.62                 | -5.53           |
| 16:00        | 8.84             | 5.86                  | 2.98            |
| 17:00        | 8.41             | 0.80                  | 7.61            |
| 18:00        | 8.02             | 0.0                   | 8.02            |
| 19:00        | 8.21             | 0.0                   | 8.21            |
| 20:00        | 7.29             | 0.0                   | 7.29            |
| 21:00        | 6.62             | 0.0                   | 6.62            |
| 22:00        | 6.81             | 0.0                   | 6.81            |
| 23:00        | 6.20             | 0.0                   | 6.20            |
| <b>Total</b> | <b>191.7</b>     | <b>58.8</b>           | —               |

*Note:* Negative net demand indicates surplus solar energy available for EV charging.

## B.3 Individual EV Trajectories

### B.3.1 EV State of Charge Evolution

Table B.5: Individual EV SoC trajectories (RL method, charge-only)

| Hour  | EV 1 | EV 2 | EV 3 | ... | EV 15 | Average |
|-------|------|------|------|-----|-------|---------|
| 00:00 | 25%  | 22%  | 28%  | ... | 24%   | 25%     |
| 06:00 | 25%  | 22%  | 28%  | ... | 24%   | 25%     |
| 07:00 | 30%  | 28%  | 33%  | ... | 29%   | 30%     |
| 08:00 | 38%  | 35%  | 41%  | ... | 37%   | 38%     |
| 09:00 | 46%  | 43%  | 49%  | ... | 45%   | 46%     |
| 10:00 | 54%  | 51%  | 57%  | ... | 53%   | 54%     |
| 11:00 | 62%  | 59%  | 65%  | ... | 61%   | 62%     |
| 12:00 | 70%  | 67%  | 73%  | ... | 69%   | 70%     |
| 13:00 | 75%  | 72%  | 78%  | ... | 74%   | 75%     |
| 14:00 | 76%  | 73%  | 79%  | ... | 75%   | 76%     |
| 15:00 | 76%  | 73%  | 79%  | ... | 75%   | 76%     |
| 16:00 | 76%  | 73%  | 79%  | ... | 75%   | 76%     |